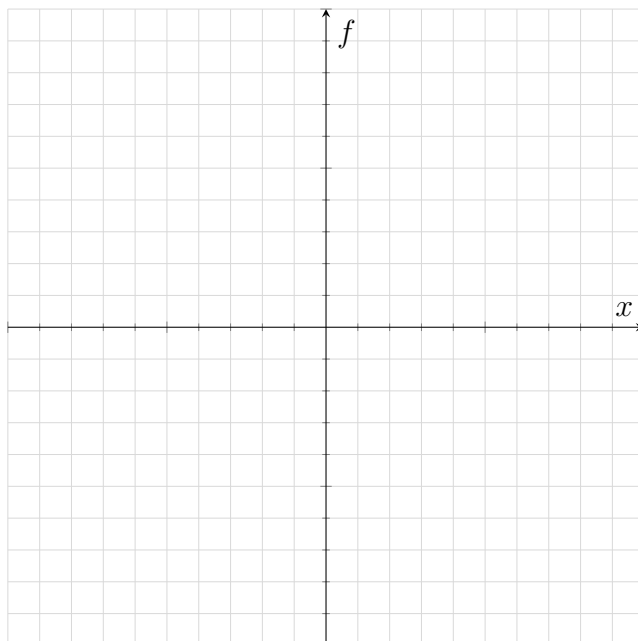


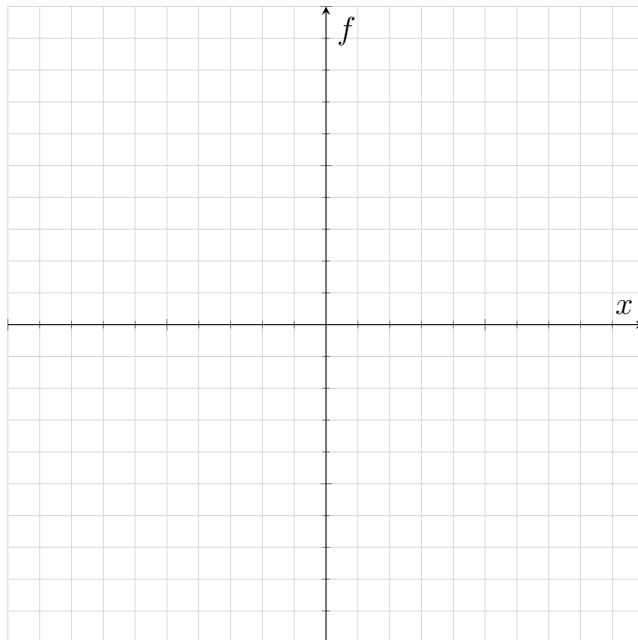
All questions are doable by hand :)

1. Plot the graph of $f(x) = \left| \frac{2}{x(x-1)(x+1)} \right|$



2. Let $f(x) = (x - a)^2|x - a| + 3|x - a| - 3(x - a)^2 + 4$. Find values of a such that the corner of f lies on the graph of $y = (x + 2)^2$

3. Sketch the graph of $f(x) = \frac{x^3 - 3x^2 + 2}{x^2 - 4}$. Label all axes intercepts and asymptotes



4. Let $a \in \mathbb{R}$, and

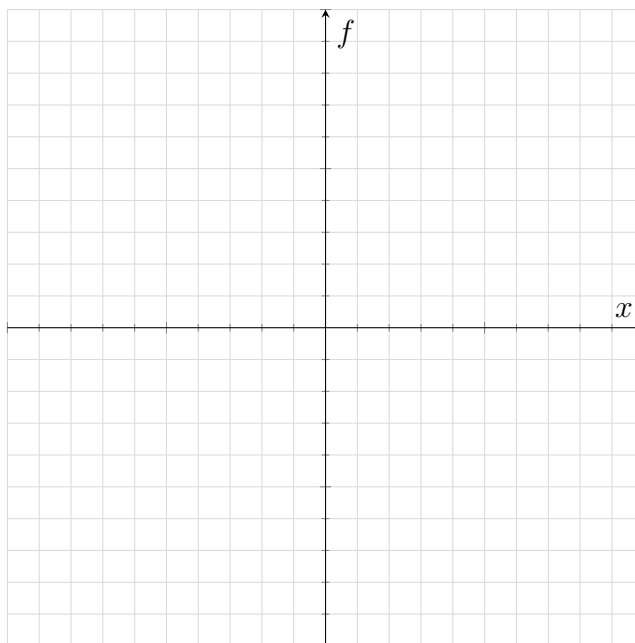
$$f : D \rightarrow \mathbb{R}, \quad f(x) = \frac{x^3 + 5x^2 - 2x - 24}{(x - a)(x - 3)}$$

- (a) find the values of a such that f has at most one vertical asymptote

- (b) Suppose $a \in (-3, 0)$.

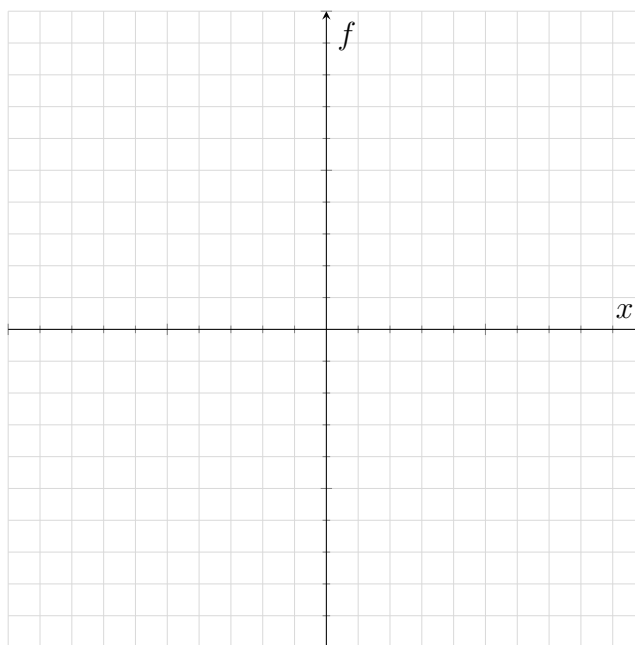
- i. Find the implied domain of f

- ii. Plot $f(x)$, labelling the axis intercepts and vertical asymptotes (You dont need to worry about **labelling** the other key features)



- (c) Plot the graph of the following. Label the axis intercepts, and any asymptotes.

$$\left| \frac{x^2|x| + 5x^2 - 2|x| - 24}{|x|(|x| - 3)} \right|$$



5. Find values of i, j such that the graph of $y = ||x - i| - j|$ has a corner at $(1, 0)$